

Student number:

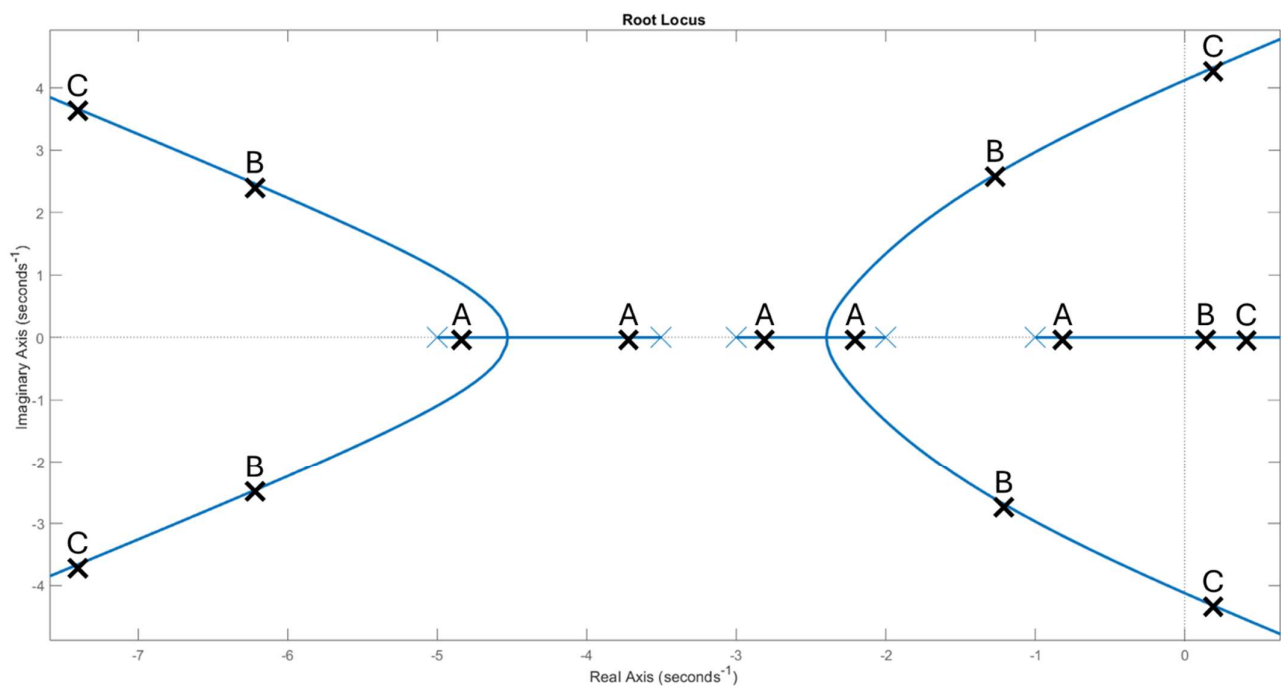
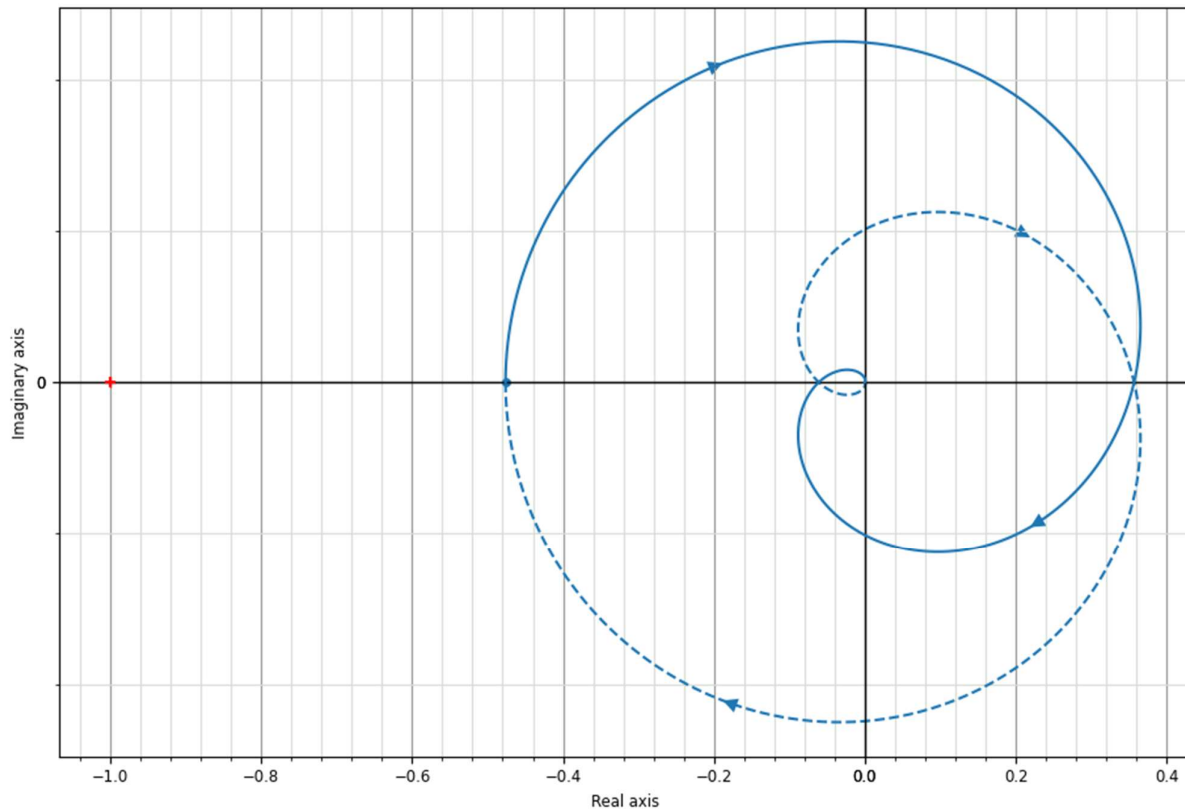


EEE3094S 2025: Class Test 2

- The test is open book. You do not need to hand in your resources.
- **The supporting information for all questions and formula sheet are attached at the end of this paper. You may remove them for easy access.**
- Answer on the test sheet. If you run out of space, you may attach additional pages, but please note that you have done so (e.g. write “see attached pages” by the question) and make sure they are marked with your student number.
- You may do your diagrams in pencil but please ensure that your lines are clear.
- There are 4 questions worth a total of 50 marks.
- You have 90 minutes to complete the test + 5 minutes of reading time.

Question 1 [9 marks]

The Nyquist plot and root locus of a system are shown below. The system is going to be placed in a negative feedback loop with a proportional controller. A few possible closed loop pole locations labelled A-C are marked on the root locus. For each one, use the Nyquist plot to determine an approximate range ("smaller than {value}", "larger than {value} but smaller than {value}", or "larger than {value}") that the gain of the proportional controller could fall into if those are the closed-loop poles. Explain your reasoning. [3 x 3 marks]



Solutions:

A) **Less than 2.08**

In this closed loop pole configuration, no poles are in the right half plane. Since there are no unstable open loop poles, the Nyquist stability criterion indicates that there should be no encirclements of -1 ($N = P_{CL} - P_{OL} = 0 - 0 = 0$).

The intersection of the negative real axis that is furthest to the left is at around -0.48, so the gain of the proportional controller would need to be at least 2.08 for this to pass -1, meaning the gain must be less than this value.

B) **Greater than 2.08 but less than 16.7**

In configuration B, there is one unstable closed-loop pole, so we would expect one clockwise encirclement. This will occur when the gain is large enough that the intersection considered in part (A) crosses -1, but too small for the intersection closer to the origin to pass -1, as this would result in 3 clockwise encirclements, and hence, three unstable closed-loop poles.

The second intersection is at between -0.04 and -0.08, so let's call it -0.06. The gain would therefore need to be around 16.7 for the point to pass -1.

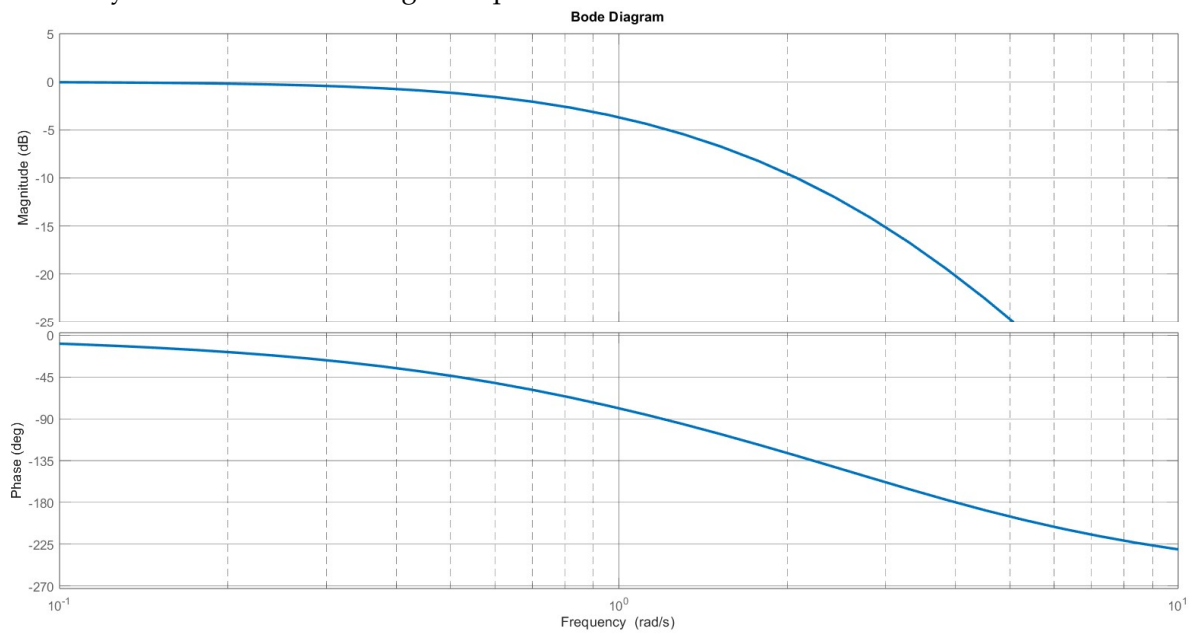
C) **Greater than 16.7**

There are now three unstable closed-loop poles, so the gain must be large enough for both intersections of the negative real axis to have passed -1. Based on the calculations in previous questions, this must mean that the gain is at least 16.7.

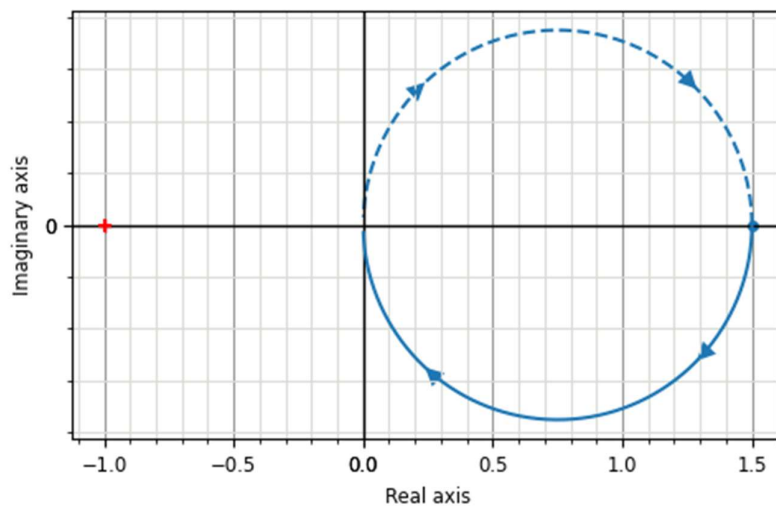
Question 2 [12 marks]

For each of the open-loop systems described, decide whether the system is stable in closed loop at its present value of gain, and if so, determine the gain margin of the system. Explain your answers.
[4 x 3 marks]

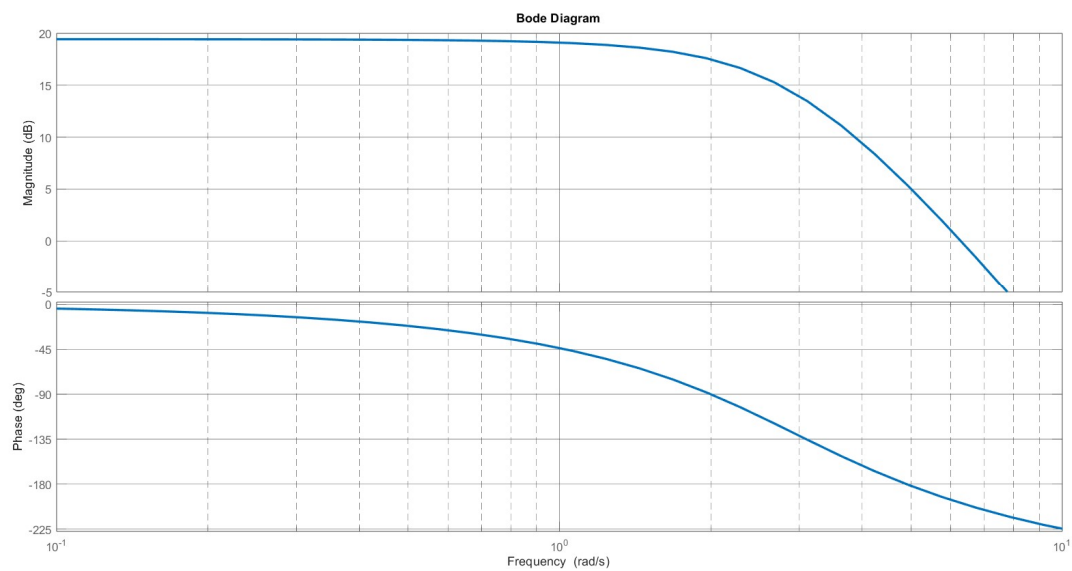
A) A stable system with the following Bode plot:



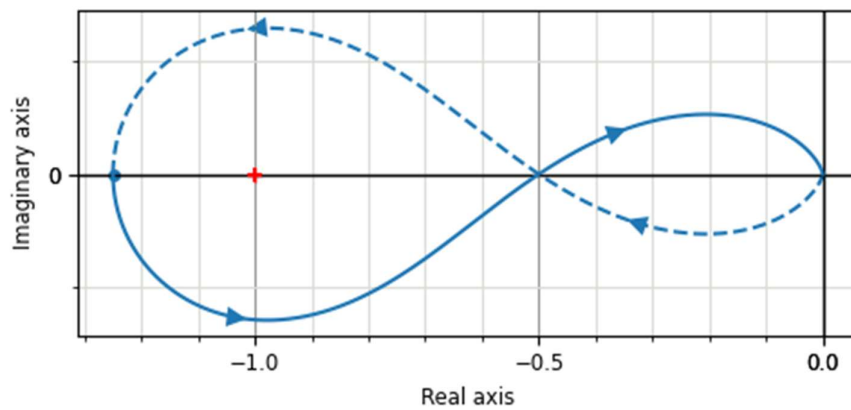
B) A stable system with the following Nyquist plot:



C) A stable system with the following Bode plot:



D) A system with one unstable open-loop pole and the following Nyquist plot:



Solutions:

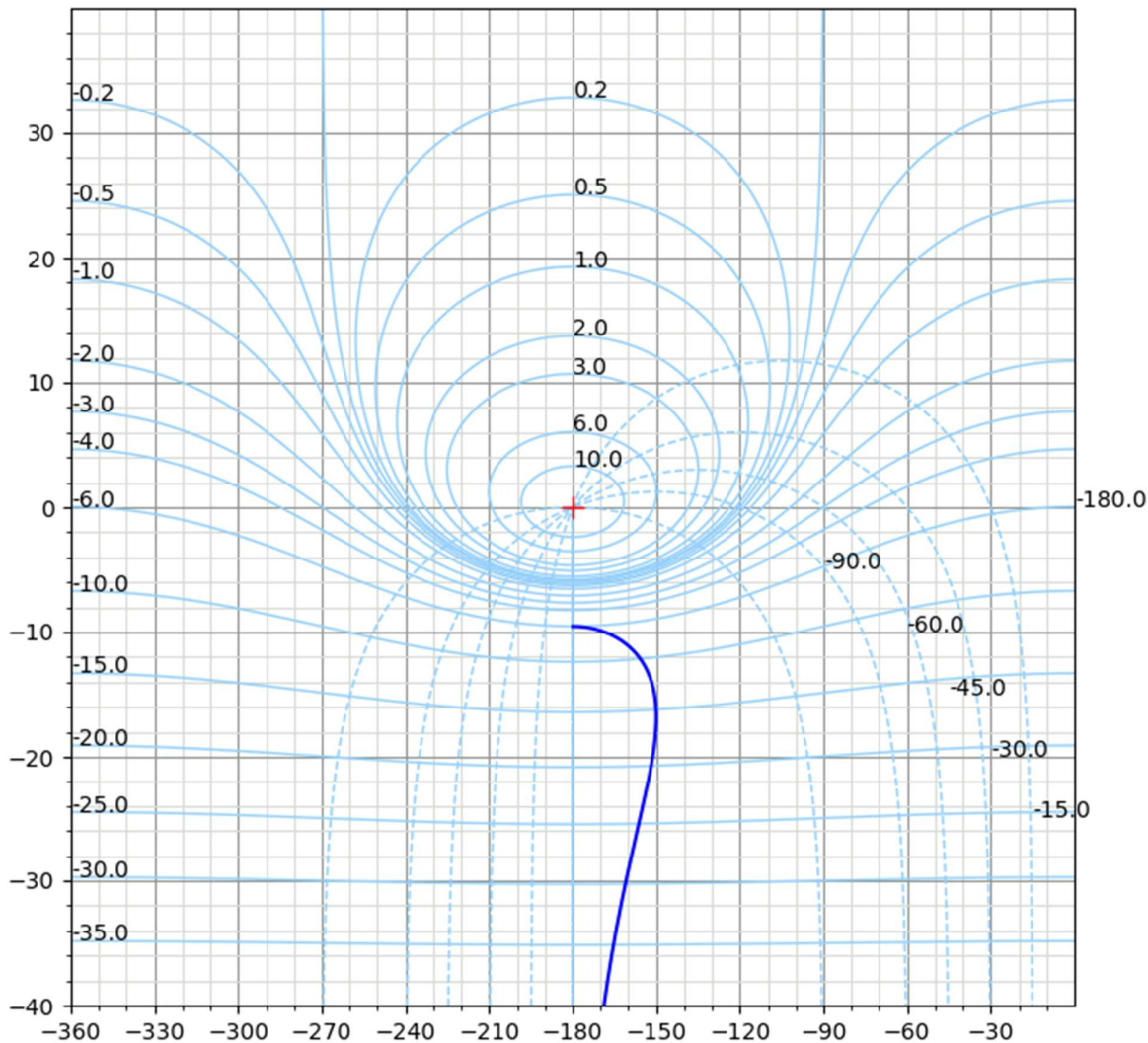
- A) The phase crossover frequency (at -180 degrees) occurs at around 4 rad/s. The gain at this value is about -20 dB (0.1), so the system is stable in closed loop with a gain margin of 10.
- B) The system has no unstable open-loop poles, so it requires zero encirclements to have zero unstable closed-loop poles. This is true at the current gain value, so the system is stable in closed loop, and it will remain true at all gains, as the plot does not intersect the negative real axis, giving infinite gain margin.
- C) The phase crossover frequency (at -180 degrees) is around 5 rad/s. The gain here is about 5 dB, which is greater than 0 dB (1), so the system is unstable in closed loop.
- D) This system has one unstable open-loop pole and one counterclockwise encirclement of -1, so the Nyquist stability criterion has $-1 = P_{CL} - 1 \therefore P_{CL} = 0$, indicating no unstable closed-loop poles.

If the gain is increased so that the intersection at 0.5 passes -1, the plot will instead have one clockwise encirclement of -1, giving $1 = P_{CL} - 1 \therefore P_{CL} = 2$ unstable poles. This will occur at a gain of 2, so this is the gain margin.

Question 3 [13 marks]

The Nichols chart of the unstable system $G(s) = \frac{1}{(s-1)(s+3)}$ is shown below.

- A) With reference to the Nichols chart, explain how you know it is possible to stabilize the system by placing it in negative feedback with a proportional controller, and determine the range of controller gains for which the closed-loop system will be stable. Indicate this on the Nichols chart by drawing a specification boundary. [4 marks]
- B) Draw specification boundaries on the plot representing the following:
- A phase margin of 30 degrees [1 mark]
 - Tracking accuracy of 90% with respect to constant setpoints. [2 marks]
- C) Is it possible to meet both the specifications in part B at once? You may draw on the plot to support your explanation. [3 marks]
- D) If you prioritize the accuracy specification in part B, ignoring all other specifications, give a best-case approximation for the percentage overshoot of the closed loop system. [3 marks]



Solutions:

- A) From the Nichols chart, we see that increasing the gain will cause the response to encircle -1 in a clockwise direction. This corresponds to an anticlockwise encirclement on the Nyquist plot, that is $N = -1$.
[1 mark]

Since the system has one unstable pole, the Nyquist stability criterion $N(-1) = P_{CL}(0) - P_{OL}(1)$ indicates that this encirclement corresponds to zero unstable closed-loop poles. Therefore, increasing the gain will stabilize the system. [1 mark]

The gain will need to be increased by around 10 dB (3) for this to occur. [1 mark] (Values between 9 and 10 dB also acceptable)

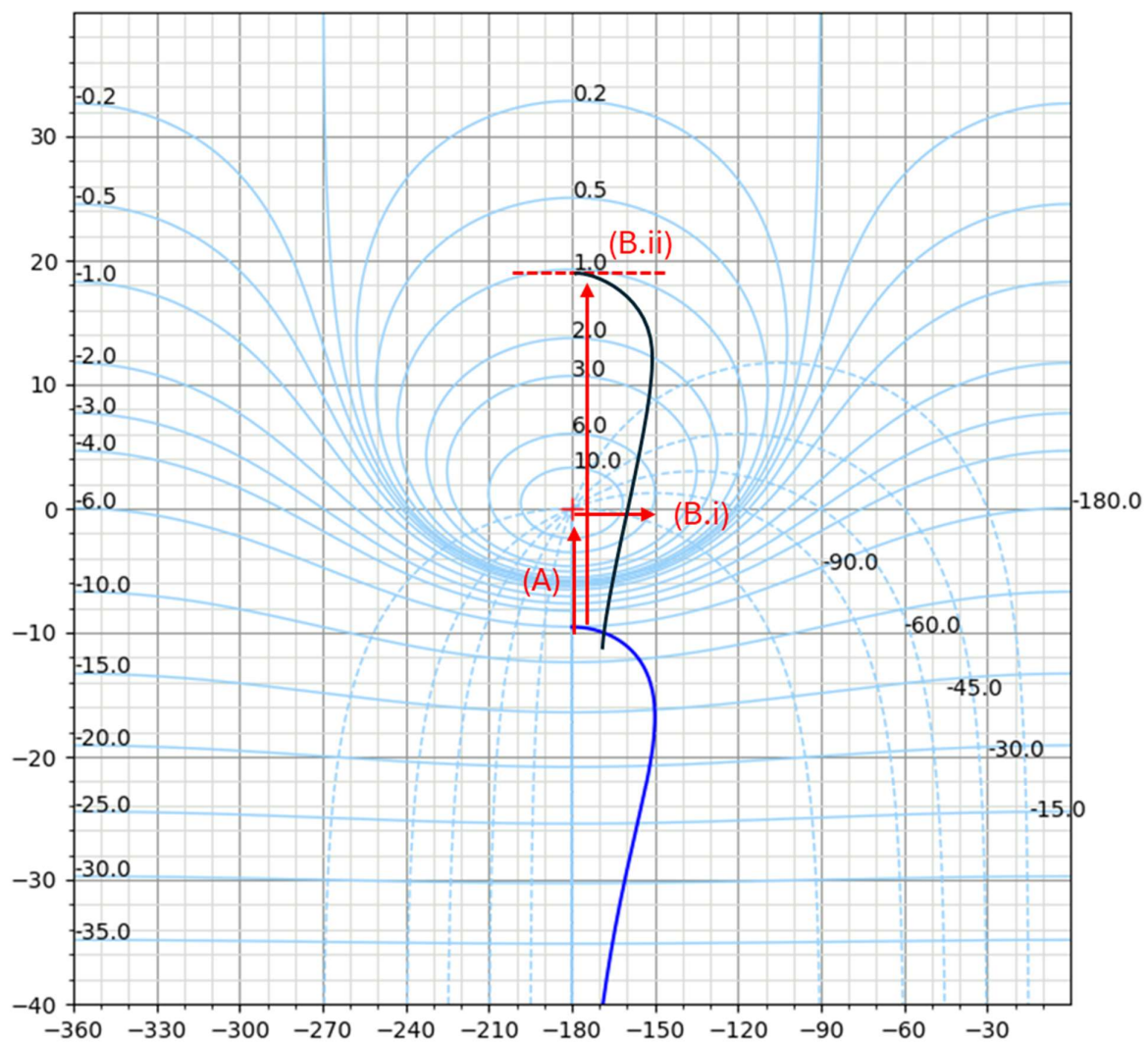
The gain margin is shown on the plot [1 mark]

- B)
- i) Shown on plot. [1 mark]
 - ii) From the final value theorem of the error, tracking with 90% accuracy (i.e. less than 10% error) means $\frac{1}{1+KG(0)} \leq 0.1 \therefore KG(0) \geq 9$, i.e. 19 dB. [1 mark]
Specification on the plot [1 mark]
- C) If we redraw the response at the gain required for the accuracy requirement [1 mark], the phase margin specification is violated, so it is not possible to meet both specifications [2 marks].
- D) The redrawn response appears to have a phase margin of around 20 degrees [1 mark].

Note that, because of the right-half plane pole, this system's steady-state error goes in the opposite direction i.e. the response will be 10% greater than the intended value, rather than 10% less. This can be seen because the low frequencies in the response with the increased gain intersect the 1 dB M circle, indicating a closed-loop gain of around 1.1 at low frequencies.

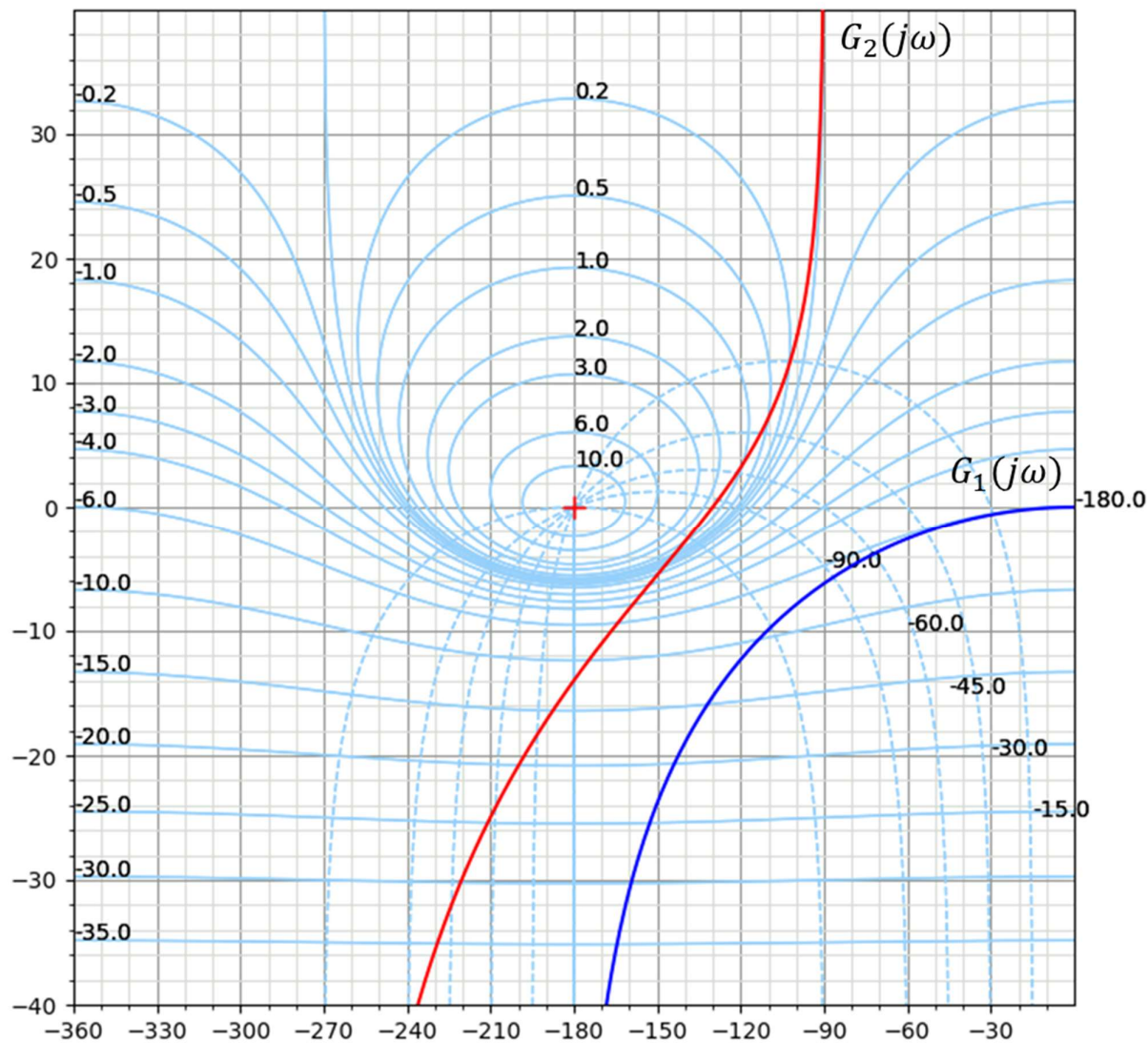
Taking this closed-loop gain into account, the phase margin indicates a damping ratio of around $\zeta = \frac{1.1}{100} 20 = 0.22$, which would give a percentage overshoot of 49.4%. [2 marks]

It is understandable to miss this detail and just go with a closed-loop DC gain of 0.9 considering the accuracy specification, so you can also get [1 mark] for $\zeta = \frac{0.9}{100} 20 = 0.18$, which corresponds to a percentage overshoot of 55.7% [1 mark]



Question 4 [14 marks]

The frequency response of a system $G_1(s)$ is shown on the Nichols chart below. This system is to be placed in negative feedback with an integral controller of the form $\frac{K}{s}$. The open-loop frequency response of the system in cascade with an integrator $\frac{1}{s}$ is shown on the same plot as $G_2(j\omega)$.



- A) Does the addition of the integrator affect the gain margin of the system? Explain. [3 marks]
- B) The aim of placing the system in negative feedback is for it to satisfy the following specifications:
- Tracking position setpoints with accuracy of more than 95%.
 - Percentage overshoot of less than 20%.
- Decide whether the system with the integral controller meets each of these specifications, supporting your answers with features from the Nichols chart. [2 × 3 marks]
- C) Your friend thinks it's unnecessary to go through the effort of implementing an integral controller, when it's possible to achieve the required tracking accuracy with a proportional controller.
- Determine the minimum gain required of this proportional controller, and re-draw the response of the original system $G_1(j\omega)$ at this new gain. [2 marks]
 - Based on the response of the system with the proportional controller, would you agree with your

friend that proportional control is the way to go? You may draw on/ annotate the Nichols chart further to support your answer. [3 marks]

Solutions:

A) Yes. The original system has an infinite gain margin, while the system with the integrator has a gain margin of around 14 dB.

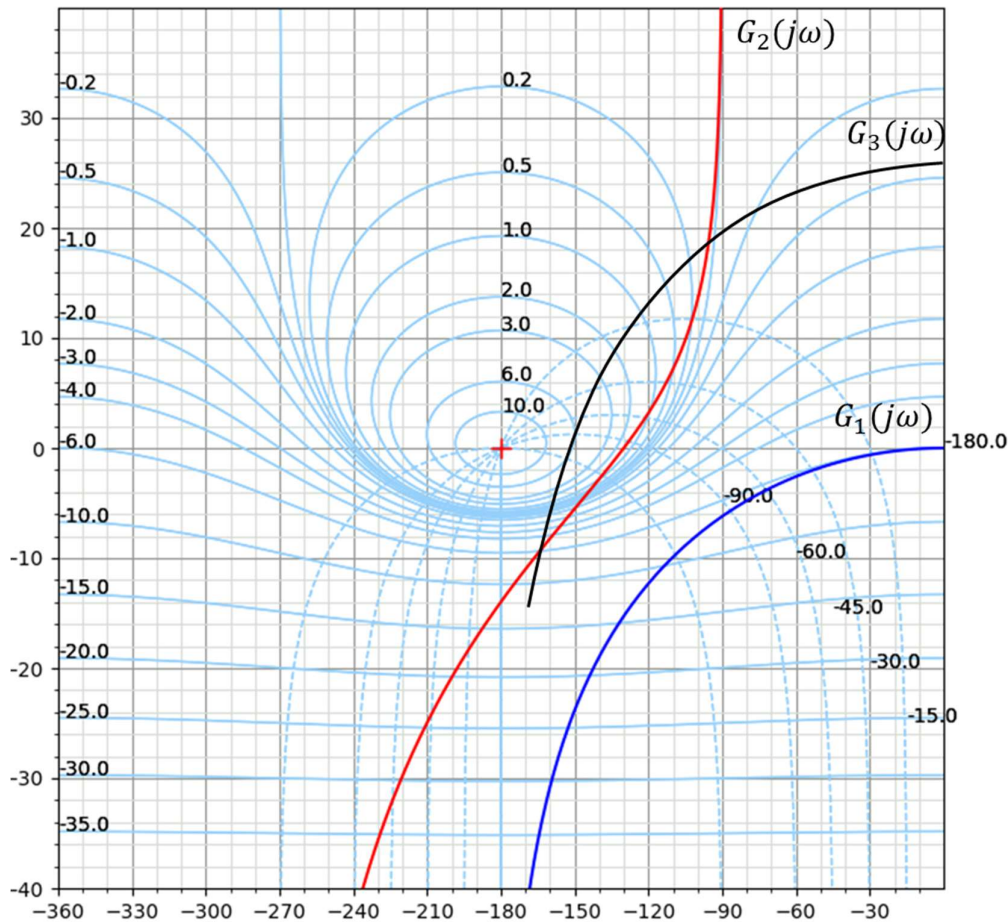
B) i) The integral controller will be able to track constant setpoints with 100% accuracy, as it raises the type number of the system to 1. This answer is sufficient, but an alternative that specifically references the Nichols chart is that the low frequencies of the response intersect the 0 dB circle, showing that the system has a gain of 1 in closed loop.

ii) The system with the integral controller has a phase margin of around 50 degrees. Combined with the closed-loop gain of 1, this means it has a closed-loop damping ratio of approximately

$$\frac{1}{100} \cdot 50 = 0.5.$$

Subbing this into the percentage overshoot formula gives a 16.3% overshoot, which is within the specifications.

C) i) The original system has a gain of 1. To track a position with 95% accuracy, $\frac{KG_1(0)}{1+KG_1(0)} \geq 0.95$, so the gain would need to be at least 19, or around 26 dB. Raising the gain to 26 dB gives the response shown in black as $G_3(j\omega)$ below:



ii) The proportionally controlled system has a phase margin of less than 30 degrees, meaning the damping ratio is less than $\frac{0.95}{100} 30 = 0.285$. This gives a percentage overshoot of more than 39%, which does not satisfy the requirements of the problem. We should therefore still use the integral controller.